

Amplifier-based MOS Analog Neural Network Implementation and Weights Optimization

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ABSTRACT

Neural networks are achieving state-of-the-art performance in many applications, from speech recognition to computer vision. A neuron in a multi-layer network needs to multiply each input by its weight, sum the results and perform an activation function. This paper presents a variation of the implementation of an amplifier-based MOS analog neuron capable of performing these tasks and also the optimization of the synaptic weights using in-loop circuit simulations. MOS transistors operating in the triode region are used as variable resistors to convert the input and weight voltage to a proportional input current. To test the analog neuron in full networks, an automatic generator is developed to produce a netlist based on the number of neurons on each layer, inputs and weights. Simulation results using a CMOS 180 nm technology demonstrate the neuron proper transfer function and its functionality while trained in test datasets.

KEYWORDS

analog neural networks, analog circuits, optimization, neuromorphic

1 INTRODUCTION

Artificial Neural Networks (ANNs) are computational models based on biological neural networks and often tailored to solve engineering problems. This field is currently under intense scientific and industrial investigation and has an increasing number of techniques and applications, such as in medical diagnose, robot control, analysis and prediction of complex systems, image and speech recognition, financial data analysis and others.

Most of current commercial applications of ANNs use software-based implementations, not taking full advantage of the parallelism allowed by neural network topologies. In the scientific literature, however, many hardware implementations were developed with different design approaches to achieve greater speed and reduced power and cost [1]. The design of such hardware implementations can be analog, digital or mixed-signal. This paper explores a MOS analog implementation, as this type of implementation can directly use the physics of MOS devices to perform the synapses.

A MOS topology for synaptic weights implementation was introduced by Salam et al. in [2], from which the implemention in this paper is a variation. In [3], a 2-neuron network was developed to demonstrate its functionality and in [4] a modified learning rule is applied to find the weights. In continuation of their previous work, Fathi and Salam [5] implemented a level-shifting in the output of the neuron, but considered the weight voltages to be in the drain of the transistor and the input voltages to be in the gate, which resulted in different requirements and implementation to the level-shifting used in this paper.

Most analog neurons are designed aiming Feedforward Multilayer Neural Networks. In [6], a 50-neuron analog chip with all-MOS implementation of artificial neuron was proposed. A similar analog reconfigurable artificial neuron was proposed in [7] using switches to determine the connections of the neuron network topology. A different approach is to use Recurrent Neural Network (RNN) design. An analog implementation of Encoded Neural Networks (a type of RNN) with computations using current-mode circuits is showed in [8].

Some works focus on application-specific analog neuron implementations. Laurent Gatet et al. [9] proposed an integrated analog neuron to be used in an real-time surface classification application. They used a modified Gilbert cell to perform the multiplication necessary to the synapses. The activation function was an hyperbolic tangent cell. In [10], a hardware neuron implementation tailored to a power amplifier predistorter application was designed. In [11], a perceptron for applications in amperometry was proposed.

This paper implements an amplifier-based analog neuron using MOS transistors operating in triode region to perform the neuron synapses. The main contributions of this paper are :

- implementation of a variation of the analog neuron in [2] with a second stage comprised of a level shifter to better cope with output voltage range and non-linearity added by diode-connected MOS transistors;
- optimization of synaptic weights using a netlist generator and in-loop circuit simulations
- study of the voltage ranges for inputs/outputs and weights for the used topology using a CMOS 180 nm technology

2 ANALOG NEURON TOPOLOGY

The input section of the topology is a differential amplifier with current sum on both amplifier inputs. Each neuron input is connected to both amplifier inputs through MOS devices, but only one path will be active and working as a variable resistance, depending on the weight signal (positive or negative). The weights of the neuron are defined by the gate potentials of the MOS transistors. Figure 1 shows the topology for an analog artificial neuron with 2 inputs.

Voltages ranges are selected so all input transistors (MW1n, MW2n, ... MW1n, MW2n,...) are operating either in cut-off region

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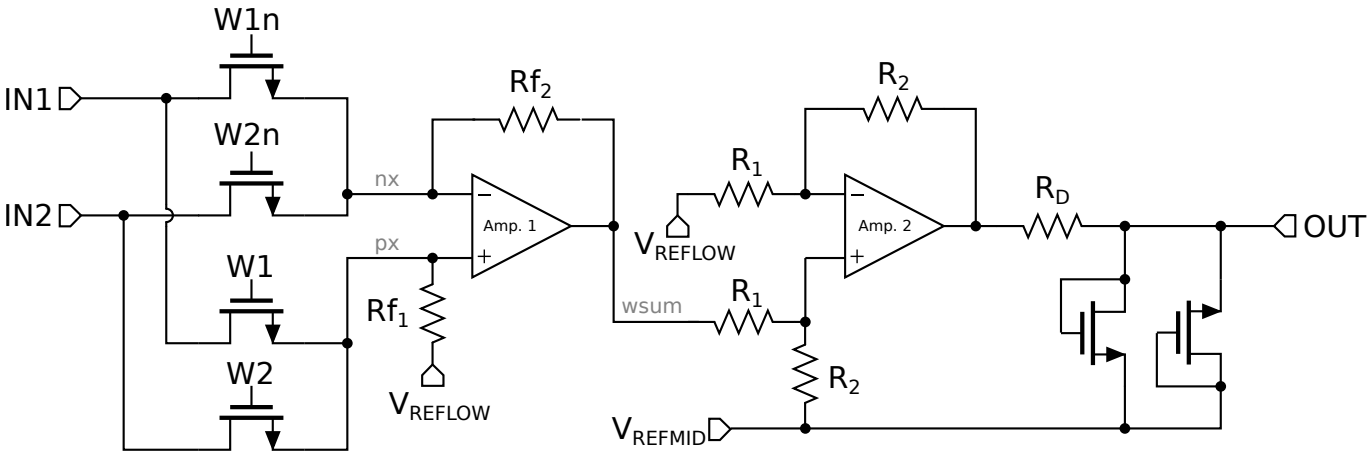


Figure 1: Two-input amplifier-based artificial analog neuron with voltage ranges adapted to allow opamp transistors to be saturated and with diode-connected transistors on the output

or in triode region. The topology can be divided in 3 parts: input weighted-sum section; level-shift and gain section; and voltage limiting section.

The input weighted-sum section is responsible for multiplying each input by its weight and sum all contributions. The reference voltage V_{REFLOW} is set to allow a voltage room for the amplifier transistors to operate in saturation.

The second section shifts the level of the output to the reference voltage V_{REFMID} and amplify its value to reach the limits of the input/output range. The last section guarantees that the output voltage will not exceed the output voltage limits and provides a non-linear shape for the analog transfer function.

2.1 Effect of Weight Voltages

Two NMOS transistors implement the impact of a weight on an input. One of them is used to provide a positive weight effect and the other a negative weight effect. Therefore, one of them will always be cut-off. Let us assume there is a variable Wv which is the actual weight of the neuron. Voltages V_{W1} and V_{W1n} will be a result of Wv as follows:

$$V_{W1} = \begin{cases} |Wv| & \text{if } Wv \geq 0 \\ 0 & otherwise \end{cases} \tag{1}$$

$$V_{W1n} = \begin{cases} 0 & \text{if } Wv \geq 0 \\ |Wv| & otherwise \end{cases} \tag{2}$$

It is desired that the same weight value, when applied to the positive or negative weight neurons, will have the same amount of impact on the output, although with opposite signal. To analyze this balance, we consider the input section of a single input neuron with one transistor for the positive weight and other for the negative weight.

Figure 2 shows a highlight on the input section of a one-input artificial analog neuron so we can better analyze the effect of the voltages on the terminal *wsum*.

For this section, let us assume the reference voltage V_{REFLOW} is equal to zero. If we consider the active transistors are in triode region, we can see the I-V relation of the transistor as a resistance:

$$R_{DS} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})} \tag{3}$$

where μ_n is the electron mobility, C_{ox} is the oxide capacitance per unit area, W is the gate width, L is the gate length, V_{GS} is the gate-source voltage and V_{TH} is the transistor threshold voltage. Considering the effect of the positive weight on *wsum*:

$$V'_{px} = V_{IN1} \cdot \frac{Rf_1}{Rf_1 + R_{M1p}} \tag{4}$$

$$V'_{wsum} = V'_{px} = V'_{nx} \tag{5}$$

Considering the effect of the negative weight on *wsum*:

$$V''_{px} = V''_{nx} = 0 \tag{6}$$

$$I''_{MW1n} = \frac{V_{IN1}}{R_{M1n}} \tag{7}$$

$$V''_{wsum} = -V_{IN1} \frac{Rf_2}{R_{M1n}} \tag{8}$$

Using $Rf_1 = Rf_2$ and guaranteeing $R_{M1p} >> R_{f1}$:

$$V_{wsum} = V'_{wsum} + V''_{wsum} \approx V_{IN1} \cdot (\frac{Rf}{R_{M1p}} - \frac{Rf}{R_{M1n}}) \tag{9}$$

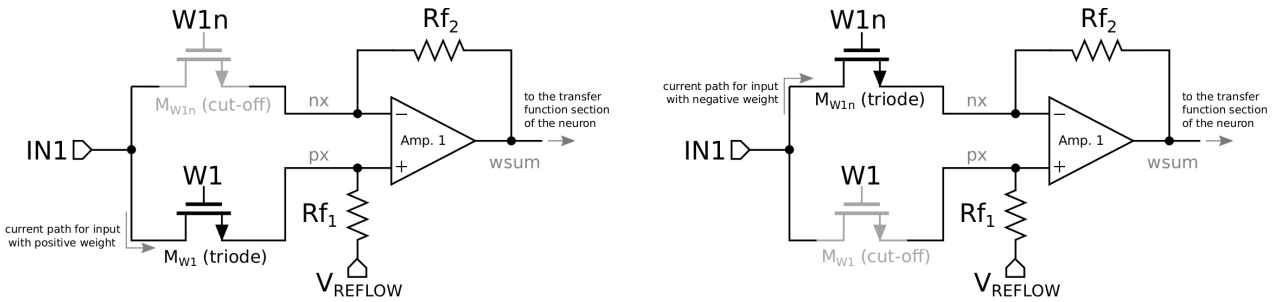
If R_{f1} is sufficiently small, V_{px} will be close to zero. In that case, if we consider a variable $KWp = \frac{Rf}{R_{M1p}}$ and $KWn = \frac{Rf}{R_{M1n}}$:

$$KWp \propto V_{W1} - V_{TH} \tag{10}$$

$$KWn \propto V_{W1n} - V_{TH} \tag{11}$$

$$V_{wsum} \approx V_{IN1} \cdot (KWp - KWn) \tag{12}$$

The same analysis will apply if we consider the reference voltage V_{REFLOW} as a low value different from zero. The output of the

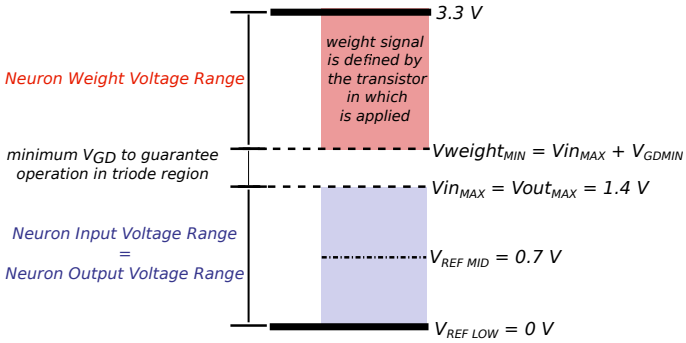


(a) Positive weight (b) Negative weight

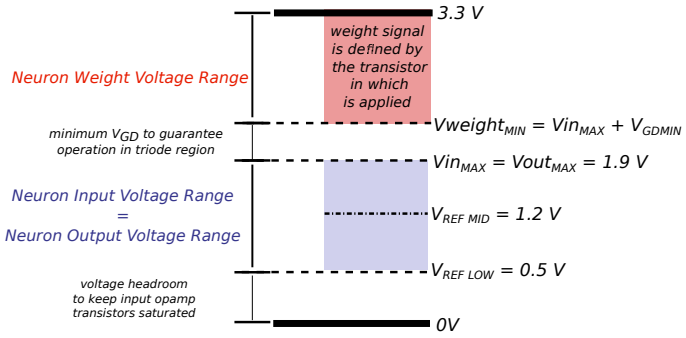
Figure 2: Input weighted sum section of a one-input artificial analog neuron

first stage will have an offset with the same value as this reference voltage.

Figure 3 shows how the voltage ranges are separated between neuron weights and neuron inputs. Figure 3 (a) shows how the voltage ranges would be selected if V_{REFLOW} had equal voltage as the VSS rail. Figure 3 (b) shows how the voltage ranges are impacted by the addition of a headroom to allow the opamp transistors to be saturated. In this work, we used the (b) set of voltage ranges. This would not be necessary if the opamp topology could reach the VSS rail. However, this was selected to simplify the opamp design.



(a) Considering V_{REFLOW} equal to the VSS rail



(b) Considering V_{REFLOW} 0.5 V greater than the VSS rail

Figure 3: Voltage ranges for the proposed analog neuron

The results presented in this work were produced using a CMOS 180 nm technology from TSMC with a VDD of 3.3V using thick-oxide transistors. Simulations were performed using ngspice with BSIM V4 models. The amplifiers were designed using a traditional PMOS-input Miller amplifier topology, as seen in figure 4.

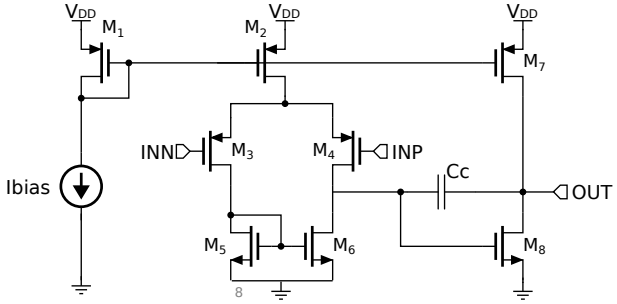


Figure 4: PMOS-input Miller amplifier topology

Figure 5 shows the effect of inputs with different weights on the output of the summing part of the neuron $V(wsum)$. Results are shown only to weight voltages in which the input transistors operate in triode region. Lower weight voltages can be used and may help the training phase, however the input transistors will be in saturation depending on neuron input voltages.

Figure 6 shows the transfer function of the second stage of the neuron.

2.2 Effect of Build-up Voltage on the Op-amp Non-Inverting Input

The voltage across resistor Rf_1 computes the sum of the contributions of the inputs with positive weights. However, this voltage is also the source terminal of all input transistors for positive weight (MW1, MW2, ...). Due to the virtual short that connects nx and px, it is also the source of all transistors responsible for negative weight (MW1n, MW2n, ...). Therefore, increases in this voltage will change V_{GS} of these transistors and, as a consequence, their R_{DS}.

This intrinsic non-linearity of the topology can be minimized by defining a small value for Rf_1. The trade-off involved is that Rf_1 should be small enough not to affect R_{DS} severely, however, the

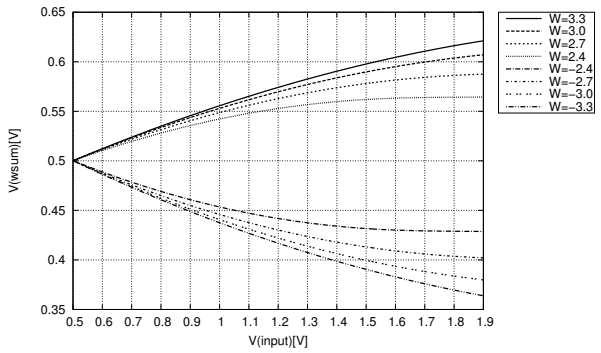


Figure 5: Effect of the input weight on the weighted sum node

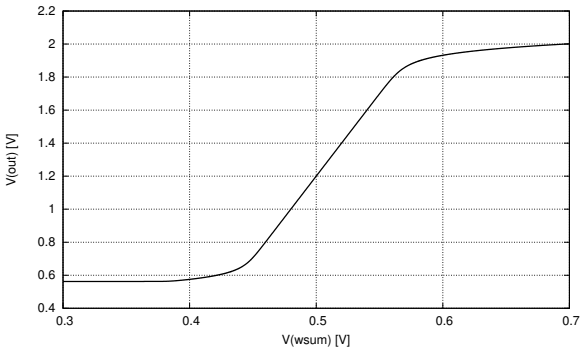


Figure 6: Transfer function (from the weighted sum node to output) of the artificial analog neuron

smaller the resistor is, more gain will have to be added to the second stage (amplifier 2). As R_{DS} is inversely proportional to $(V_{GS} - V_{TH})$, $\Delta V_S << V_G$ is a condition to produce a linear circuit. In training tests, however, even using a R_{DS} of $10k\Omega$, this non-linearity did not represent a problem for convergence of the weights to a solution with low error, showing the robustness of neural networks.

3 NETLISTS GENERATION AND TRAINING (WEIGHTS OPTIMIZATION)

To test the neuron in complete neuron networks and datasets, we developed a program to generate the circuit netlists, test and optimize the weights.

The program is implemented in python and uses the Ngspice electrical simulator in-loop to assess each solution (set of candidate weights). The designer has the option to simulate using the network considering ideal opamps or the Miller op-amp. To decrease the simulation time during training, these simulations used ideal opamps as the important in this phase is to achieve the best weights.

Figure 7 shows the flow of the program to find the weights using the electrical simulator in an optimization loop.

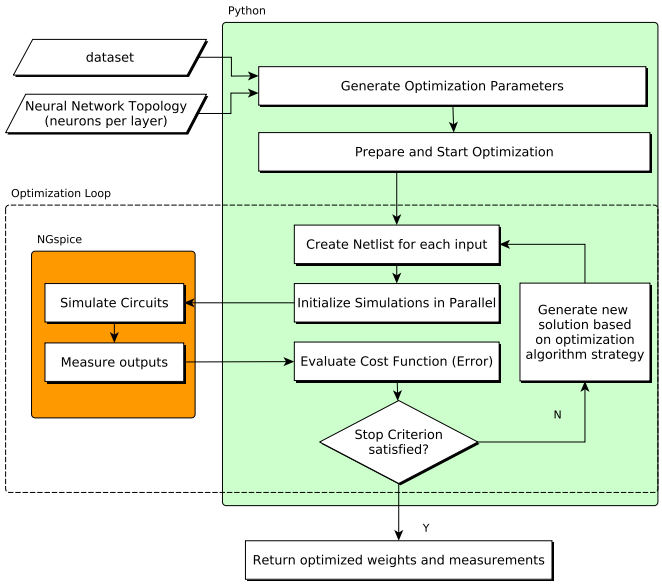


Figure 7: Program flow.

A derivative-free algorithm was used to the weight optimization due to the robustness and flexibility of this type of approach. The use of derivative-free algorithms together with an electrical simulator reduce the modelling effort required to properly represent the system and increase the accuracy of the results.

The optimization algorithm is a variation of Simulated Annealing [12], which uses a temperature parameter to avoid local minima. After every 30 iterations, the temperature parameter decreases to 90% of its previous value. Restarts at the best known solution to the moment occur every time more than 25 new solutions are generated without any improvement. In such restarts, the temperature rises to its original value (100 degrees).

4 EXPERIMENTS AND RESULTS

The tests in this section involve two training problems. The first aims only at assessing the circuit to act as a neuron, consisting of a XOR logic gate training test. The second uses the well-known dataset of iris plant classification [13].

The training for each set was performed using an electrical simulator (Ngspice) in loop with an optimization algorithm (a variation of Simulated Annealing). Computations were performed using an 8-core Intel(R) Core(TM) i7-8550U CPU @ 1.80GHz.

The stop criteria for the training phase of the weights was 3000 iterations or achieving a mean square error inferior than 0.05.

For the experiments, weight voltages were allowed to go from 1.9 to 3.3V. This results in input transistors operating in saturation region for part of the input range. However, better results can be obtained in the training phase with this extended range, as for part of the input range the transistors will be in the triode region and more weight impact will be available.

4.1 XOR testbench

The XOR function is tested initially to verify the neural network functionality and the training phase. Although it is a simple function to model, it is non-linear and useful to assess the basic functionality of the neuron in a small network. Figure 8 shows the network topology, using 3 neurons in the hidden-layer and 1 in the output. Each arrow has an weight associated with it, and b1, b2, b3 and b4 are the bias of each neuron.

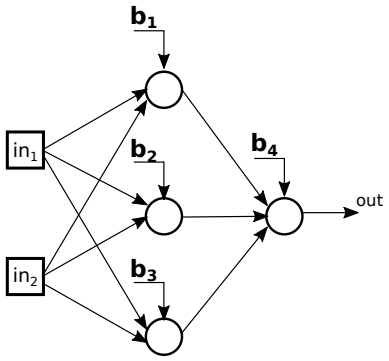


Figure 8: Network topology used for XOR training

The training phase required 1560 iterations and took 1 minute and 56 seconds. Figure 9 shows the error for each iteration during training.

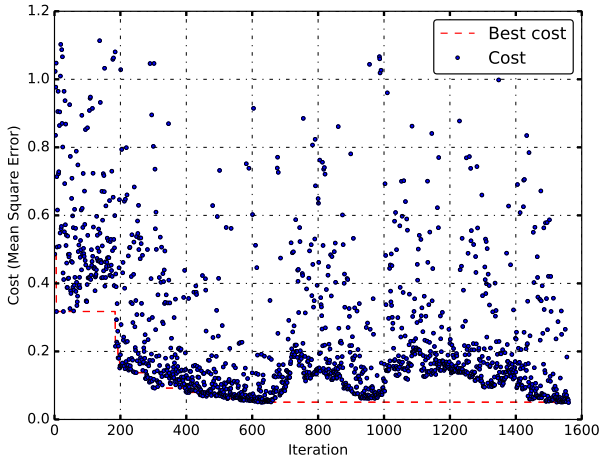


Figure 9: XOR training error for each iteration

Table 1 shows the results of the trained circuit, which indicate the circuit responds correctly for all inputs.

4.2 Iris Plant Classification

The iris plant classification problem [13] is vastly used in AI learning, specially to test new strategies and compare results. It is a simple domain in which three classes of plant (Iris Setosa, Iris Versicolour and Iris Virginica) have to identified based on four parameters (sepal length, sepal width, petal length and petal width). The dataset was divided in 80% for training and 20% for testing.

Input Voltage [V]	Expected Output [V]	Voltage Output* [V]	Output class
[0.5, 0.5]	0.5	0.68	0
[0.5, 1.9]	1.9	1.84	1
[1.9, 0.5]	1.9	1.62	1
[1.9, 1.9]	0.5	0.73	0

* Closer voltages to the expected outputs can be achieved by the addition of extra neurons in the hidden layer. In this test, however, the objective was to have a small number of neurons to prove functionality.

Table 1: XOR results using non-ideal models for the opamps (PMOS-input Miller opamps in each neuron)

Figure 10 show the network topology, using 6 neurons in the hidden-layer and 3 in the output (one for each class).

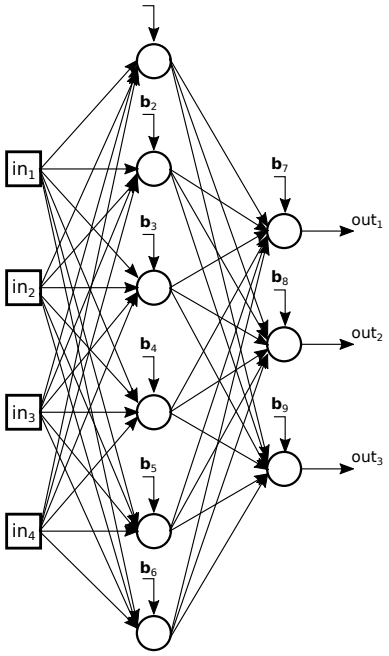


Figure 10: Network network configuration used for Iris dataset training

The training phase stopped after 3000 iterations (stop criterion) and took 1 hour and 31 minutes, mainly due to the use of electrical simulations in the loop. Figure 11 shows the error for each iteration during training.

Although the design and training show the capabilities of this analog neural network approach, it was not optimized for low power consumption, which is mostly related to the amplifier design trade-offs. The Miller amplifier used was optimized to provide a relatively low output impedance, which resulted in higher current consumption. The total power of the iris classification circuit (18 amplifiers) was 39 mW with a settling time inferior to 5 μs.

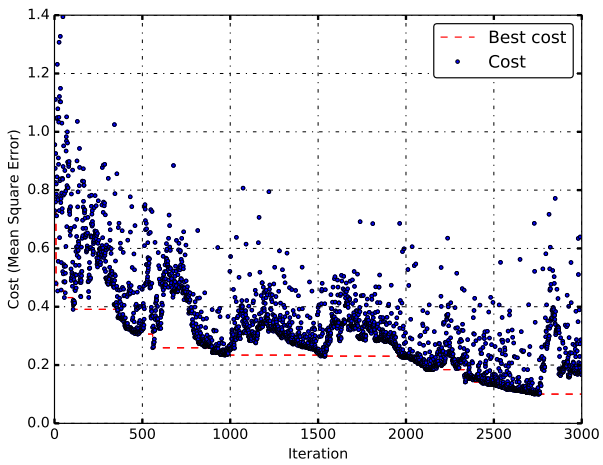


Figure 11: Iris dataset training error for each iteration

Dataset	XOR	Iris dataset
Number of Neurons	4	9
Neurons per layer	[3 1]	[6 3]
Training time	116 seconds	5447 seconds
Training accuracy	100.0%	93.3%
Test accuracy	- *	96.7%

* the XOR case did not need test dataset as all possible cases are already in the training dataset

Table 2: Results for XOR and Iris dataset

5 CONCLUSION

An amplifier-based artificial analog neuron was presented, similar to previous approaches as in [2] and [3], but with a second stage comprised of a level shifter to better cope with output voltage range and non-linearity added by diode-connected MOS transistors. The neuron uses input transistors operating in the triode region and has a separate signal path for positive and negative weights.

The artificial neuron was used as a building block to create neural networks with the TSMC 180 nm technology. After training, simulation results showed that, both for a simple XOR testbench and for an Iris Plant Classification dataset, the accuracy of the final networks indicate the proper functionality of the neuron and the networks.

The voltage range available to represent neuron weights in a 180nm technology using this topology is sufficient to properly achieve good results with a small number of neurons. However, the use of two amplifiers per neuron will result in high power consumption if a large network is needed. Therefore, we consider the results promising but further improvements such as reducing the number of components are needed to allow similar designs to be easily scalable to large networks.

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